

Experimental Methods

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Economics: an observational science

- Economic theories are devised to explain market activity.
 - Technically sophisticated models
 - But limited capacity to evaluate their predictive content
- Game theoretic predictions depend on subtle behavioral assumptions and fine institutional detail.
- Extremely difficult to test abstraction assumptions in naturally occurring markets.

Laboratory Experiments

- Other observational sciences overcome the obstacles inherent in field data by systematically collecting data in controlled, laboratory conditions.
- For example, Astronomy is founded on
 - Propositions for particle physics
 - Propositions painstakingly evaluated in the laboratory.
- Use of laboratory experiments has become wide spread in economics.

Experimental Methods: Advantages

- Replicability
 - Other investigators can reproduce the experiment
 - Verify the findings independently.
 - Provides professional incentives to carefully collect relevant data

Contrast with economic field data

- Economic field data
 - Not replicable
 - Numerous unobserved factors are constantly changing.
 - Not collected for scientific purposes
 - Often difficult to verify accuracy

Experimental Method: Advantages

- Control
 - Manipulate laboratory conditions to evaluate alternative theories and policies.
 - Taking an action to affect the value of a variable
 - For example, induce value by paying money.
 - Reduces the number of auxiliary hypotheses involved in testing a primary hypothesis.
- Control is usually missing from field settings.
 - For example, subtle game theoretic refinements difficult to test with nonexperimental data.
 - Natural markets are hard to evaluate without data on supply and demand functions.

Experimental Method: Limitations

- Subject pool Sophistication
 - Real traders are more sophisticated than typical subject pool of undergraduate students.
 - Nothing limits the method to undergraduates.
 - Students have been found to be better at meeting the cognitive demands of an experiment than many (most) other subject pools.

Experimental Method: Limitations

- Complexity
 - Field environments are more complicated than lab environments
 - If a theory fails to work in a simple environment, there is little reason to expect it to work in a field environment.
 - Complexity of field setting limits claims from laboratory findings about the performance of the field.

Alternatives to control

- Measurement
 - Revealed preferences
 - Measures of risk aversion using certainty equivalents
 - Measures of probability judgments using scoring rules
 - Brain scans
- Assumption
 - Experimenter maintains a hypothesis about the value of a variable.

Types of Experiments

- Tests of behavioral hypotheses
 - Theory falsification
 - An environment that satisfies as many structural assumptions of a theory as possible, gives its behavioral implications their best chance.
 - See sequential bargaining literature and the fate of the subgame perfect equilibrium prediction.
 - Rubinstein concludes “My goal as a teacher is to deliver a loud and clear message that game theoretic models are not meant to supply predictions of strategic behavior in real life.” <http://gametheory.tau.ac.il/>

Types of Experiments

- Theory stress tests
 - Is the accuracy of the theories predictions sensitive to simplifying assumptions?
 - For example, Smith (1976, p. 57) on competitive equilibrium outcomes:
 - “There are no experimental results more important ...than that the information specifications of traditional competitive price theory are grossly overstated. The experimental facts are that no double auction trader needs to know *anything* about the valuation conditions of other traders, or have *any* understanding or knowledge of market supply and demand conditions, or have *any* trading experience ... or satisfy the quaint and irrelevant requirement of being a price “taker” ...”

Types of Experiments

- Searching for empirical regularities
 - Discover surprising regularities in relationships between observed economic variables.
 - For example, certain environments with multiple equilibria nevertheless have a regular convergence to a particular equilibrium.
 - Suggests new theories
 - Strategic thinking
 - Learning
 - Quantal response equilibria

Terminology

- **Session:** a sequence of periods, games, or other decision tasks involving the same group of subjects on the same day
- **Cohort:** a group of subjects that participated in a session
- **Treatment:** a unique environment or configuration of treatment variables, for example, of information, experience, incentives, and rules
- **Cell:** a set of sessions with the same experimental treatment conditions
- **Experiment design:** a specification of sessions in one or more cells to evaluate the propositions of interest
- **Experiment:** the collection of sessions in one or more related cells.

Procedural Regularity

- The ability to replicate an experiment requires certain reporting conventions, see Palfrey and Porter (1991):
 - Instructions
 - Any illustrative examples or tests of understanding
 - Criteria for answering subjects' questions
 - The nature of the monetary or other rewards
 - The presence of practice periods with no rewards
 - The subject pool and recruiting method
 - The number and experience of subjects
 - Procedures for matching subjects and roles
 - Any intentional deception of subjects
 - Procedural irregularities in specific sessions

Maintenance of a subject pool

- Rumors about the experiment.
- Group identity
- Reputation for honest instructions
 - Deceptive experiments have an external effect on our ability to conduct laboratory experiments with the subject pool.
 - If subjects expect to be deceived can't establish common knowledge of G.
 - Some argue the externality is so great that deceptive experiments should not be published.
- Record of subject experience

Motivation

- Saliency – subject decisions have a prominent effect on rewards
 - Subjects perceive the relationship between decisions made and payoff outcomes
 - Induced rewards are high enough to matter in the sense that they dominate subjective costs of making decisions.

Motivation

- Rewards must be evaluated in context
 - Participants do well maximizing hypothetical payoffs
 - No amount of money can motivate subjects to perform calculations beyond their intellectual capacity.
 - Salient rewards reduce performance variability.
- Smith (1962) suggested that, although people may have enough intrinsic motivation to earn lots of points, hypothetical rewards were typically more “erratic, unreliable, and easily satiated.”
- Few results that disconfirm theory have been overturned by paying more money.

Motivation

- Camerer and Hogarth (1999)
 - Confirm that paying money does reduce variation and outliers
 - Paying money improved performance most reliably in judgment and decision-making tasks, when there are returns to thinking harder.
- A convention amongst experimental economics is to pay subjects enough to offset the opportunity cost of their time.

Unbiasedness

- Experiments should be conducted so as to avoid experimenter induced bias
- Human participants can be eager to do what the researcher desires
- Care needs to be exercised so that the experiment does not lead participants to perceive any particular behavioral pattern as being correct or expected
 - Examples of possibly biased terms: pollution permits, cartel, or conspiracy.
 - Use neutral language unless the explicit suggestion is a treatment variable.

Anonymity and Privacy

- Most experiments make it difficult for the subjects to know who they are interacting with
 - Avoids uncontrolled influences from outside the experiment
- Subjects are usually paid in private
 - Avoids problems of envy and benevolence.

Calibration

- Experiments often investigate comparative predictions
 - Baseline treatment
 - Treatment of interest
- Calibration requires cleanly separating theoretical predictions
 - Behavioral theories are falsified rather than validated
 - Falsification is more convincing if there is a reasonable alternative that is not falsified
 - Rejection of reasonable alternatives strengthens the failure to reject the maintained hypothesis.

Anticipated performance variability

- Factors that regularly affect performance
 - Experience with the experimental environment
 - Group effects
 - Order in which treatments are presented
- Control for order effects
 - If experiment has two treatments A and B
 - Run AB and BA
 - Include order dummy variable in the statistical analyses
- Experimental design (treatment cells, blocking, and number of trials) should be informed by anticipated statistical analysis of generated data.

Within versus between subject designs



- Within-subjects design – a single subject is observed in different treatments
 - Within subjects design are more statistically powerful
 - Subject serves as her own control group
- Between-subjects design – different subjects are used in treatment A and B
 - Avoids the positive or negative transfer of behavior from one treatment into the other.

Design Parallelism



- How close should one come to the natural situation rather than to the theories about the natural situation?
 - If purpose is stress test of theory, then closeness to the natural situation is desirable.
 - If purpose is theory falsification, then closeness to the theory is desirable.
 - Problem:
 - Since many theories are equilibrium theories, experimenter has problem of allowing learning, hence, subjects often play the game repeatedly.
 - But then folk theorem comes into play.

Design Parallelism

- Given complexity of natural situations, theorists and experimenters should be cautious when making claims about the natural situations.